



Recovery Strategies to Optimise Adaptations to Concurrent Aerobic and Strength Training

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Introduction

Concurrent training represents a broad framework whereby divergent exercise modalities (e.g. endurance and strength training) are undertaken concomitantly within a single session or across a larger training cycle [1]. Specifically, it involves the mixed use of continuous submaximal/intermittent exercise, alongside higher-load resistance exercise to stimulate metabolic and morphological adaptations associated with the differing training modalities [1, 2]. The rationale for this training method is discussed in detail elsewhere in this book, though it is used by numerous athletic events requiring both endurance and strength/power training for successful performance outcomes [2]. For example, simultaneous endurance and resistance training is common for those in sports such as cycling, running and team sports [3], as well as the general population seeking to achieve a broad range of exercise-related outcomes simultaneously [4]. However, by engaging in concurrent training, both the load (volume and intensity) and potential interference to adaptive responses will increase [5]. Consequently, there is an additional requirement for appropriate recovery strategies to promote adaptation or prevent interference and non-functional overreaching [5]. Exploration of appropriate recovery methods may also be of use to limit the potential negative outcomes of concurrent training.

Recovery is a multifaceted concept that describes the return of physiological and psychological systems to (near) pre-exercise levels [6]. Recovery is also specific to the imposed physiological disturbance, and thus selected recovery interventions should be similarly specific to imposed loads [6]. Accordingly, key issues related to the training loads and stimuli of concurrent training necessitate awareness of appropriate recovery strategies. Firstly, the cumulative effects of

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training-induced fatigue resulting from concurrent training sessions can negatively impact subsequent training bouts [7]. With this loading issue in mind, appropriate recovery methods may be deemed advantageous to improve the ability to perform and tolerate successive training bouts. Secondly, evidence suggests that acute adaptive molecular signalling processes related to endurance and resistance exercise may interact in some way, resulting in possibly antagonistic responses [8]. Thus, the optimal periodisation of these respective training stimuli may help to reduce molecular ‘crosstalk’ that can suppress phenotypic adaptations [8]. Finally, the nutritional requirements to aid recovery and promote myofibrillar and mitochondrial adaptation within concurrent training warrant consideration, particularly, training-induced glycogen depletion, timing and quantity of protein consumption, and dietary supplementation strategies [7].

Concurrent Training and the Need for Recovery

Further exploring the aforementioned potential issues, the completion of multiple training sessions in close temporal proximity, as well as accumulated fatigue from greater overall training volumes, can impact upon strength and hypertrophic adaptations [4]. Specifically, evidence suggests that localised fatigue can persist for several hours following aerobic exercise, possibly impacting on the volume of work performed during subsequent resistance exercise bouts [9]. Accordingly, it is recommended that athletes separate same-day training sessions by several hours to minimise any potential interference, and perhaps more importantly, reduce the effects of fatigue as discussed in previous chapters [7]. Ultimately, recovery interventions that promote higher-quality training of either modality are important when undertaking concurrent training within or between days [7].

A separate concern of concurrent training is the ‘crosstalk’ in molecular signalling driving adaptations from the varied training stimuli [2]. For example, it is suggested that aerobic exercise may inhibit load-induced anabolic signalling via the downregulation of hypertrophic mechanisms related to energy-sensitive signals [4]. However, despite potential for acute molecular ‘crosstalk’, it is unclear whether these responses are predictive of long-term adaptations [4], and the current data do not ultimately seem to support a chronic interference effect [10]. Nonetheless, promoting an optimal recovery state, particularly from peri-training nutritional intake, may be beneficial to optimise concurrent training adaptations and minimise potential interference [1]. Hypertrophic signalling may be affected by a number of antagonistic processes related to substrate availability, and thus the overall energy status and individual nutrient intake following exercise should be considered to aid nutrition-driven recovery [7]. For example, the importance of protein or carbohydrate availability to enhance muscle recovery warrants careful consideration [11]. In addition, many dietary supplements, including those that reduce muscle damage, optimise perceived recovery, or enhance subsequent training bouts, may be of use in the context of concurrent training. Considering the abovementioned issues, the following sections will examine specific and practical strategies to enhance post-exercise recovery.

Nutritional Considerations for Recovery

Carbohydrate

Given that both resistance and endurance exercise can decrease intramuscular and hepatic glycogen stores, carbohydrate (CHO) feeding may occupy a central role in training recovery [1, 12]. Specifically, low muscle glycogen is related to increased fatigue and may inhibit performance [13, 14] and protein synthesis signalling [4]. Thus, it is recommended to consume CHO following fatiguing training sessions to minimise blunted recovery induced by low glycogen status [15], alongside ensuring adequate substrate availability for subsequent sessions. In particular, when the between-session recovery period is <8 h, CHO should be consumed as soon as possible following training, with the required quantities varying from 3–5 g kg day⁻¹ (low-load activity) to 8–12 g kg day⁻¹ (>4 h training per day) [16]. Separately, the synergistic interaction of CHO and protein as part of a post-exercise nutritional plan suggests some potential for aiding recovery within the context of concurrent resistance and aerobic training loads [16]. However, resistance training adaptations may occur irrespective of glycogen availability [17, 18], and thus there is an increased emphasis on the role of protein during recovery for concurrent athletes.

Protein

Dietary protein is a key consideration during concurrent training given its essential role in the remodelling and recovery of skeletal muscle [1]. Furthermore, if endurance and resistance exercise are performed on the same day, a rapidly absorbed protein source may be appropriate for between-session recovery [7]. A leucine-rich protein (e.g. whey) is recommended for consumption post-resistance exercise, and approximately 0.25 g kg⁻¹ of high-quality protein is suggested to optimise post-session protein turnover and stimulate myofibrillar hypertrophy—a mechanism which can be compromised during concurrent training [1]. Also, the timing of protein intake may be of importance during concurrent training. For example, repeated doses of ~20 g high-quality protein every 3 h was more effective to induce a positive protein balance than more frequent (10 g every 90 min) or less frequent (40 g every 6 h) consumption patterns and thus bears relevance to multiple training sessions per day [19]. Furthermore, a larger dose of protein (~0.5 g kg⁻¹) may be appropriate prior to sleep to offset the catabolic effects of the overnight fast and further enhance post-exercise recovery [20].

Dietary Supplementation to Enhance Recovery

Creatine

Creatine supplementation represents a potential strategy to optimise concurrent training adaptations, particularly given the issues related to fatigue accumulation from high-volume training. Exogenous creatine improves work capacity via

accelerated resynthesis of phosphocreatine and subsequent adenosine diphosphate buffering, increasing fatigue resistance and providing a potential buffer to metabolic acidosis [21]. Nutritional supplementation using creatine monohydrate (~5 g/day) may facilitate greater adaptations to concurrent training via an increased capacity to train [1]. Specifically, creatine monohydrate supplementation attenuated strength loss during concurrent training in recreationally strength-trained individuals [22]. Conversely, no advantage from creatine supplementation was reported in rowers during a 10-week combined strength and aerobic interval training programme [23]. Evidently, a paucity of data exists from which to draw conclusions for concurrent athletes; however, given the established ergogenic effects of creatine supplementation on resistance training and the role it plays in fatigue resistance, it is a promising strategy to minimise the fatiguing effect of consecutive exercise sessions [21]. Thus, it may serve to minimise interference between training sessions and optimise the recovery process during concurrent training.

Nitrate

Present in many leafy green vegetables, dietary nitrate represents a potential ergogenic strategy to assist recovery for concurrent athletes. Under conditions of hypoxia and low pH, nitrate is reduced to nitric oxide (NO), which may subsequently confer performance-enhancing effects through regulation of peripheral blood flow, mitochondrial respiration and excitation-contraction coupling [24]. Acute administration of beetroot juice supplementation (≈ 0.1 mmol kg⁻¹ body mass NO) is proposed to lower the oxygen cost of exercise and improve exercise tolerance, particularly under hypoxic conditions [25]. The ergogenic effects of exogenous nitrate supplementation may be useful to offset fatigue-related decrements in performance that can be evident when training in a pre-fatigued state, as is often the case for athletes undertaking multiple sessions involved in concurrent training [1]. However, current evidence relates primarily to endurance events, and as such it is difficult to make recommendations for concurrent training. That said, and noting that fatigue-management strategies are a primary consideration in this scenario, a potential application exists for supplementation with nitrate in concurrent training. Questions still remain regarding its efficacy in highly trained athletes [25] and the potential negative effects of chronic use [26], though nitrate supplementation still represents a potential tool to enhance recovery. As with any nutritional supplement, it must be applied with careful consideration of dosage, timing, exercise modalities and the training status of the user.

Caffeine

Caffeine is an extensively studied and commonly used dietary supplement which is supported as having a potent performance-enhancing effect [27]. It is a rapidly absorbed compound that is readily able to cross the blood-brain barrier [28] and

works to reduce perceptions of fatigue [27]. Caffeine may exert an analgesic effect through enhanced secretion of β -endorphins [29] and increase substrate mobilisation via adrenal cortisol release [29]. Further, it is suggested to act upon the central nervous system to enhance neuromuscular function and contractile force, alongside improved alertness and reaction time [27]. In the context of concurrent training, the need to tolerate increased training volumes and manage fatiguing stimuli suggests a potential role for caffeine. Additionally, co-ingestion with CHO has shown some effectiveness to increase glycogen resynthesis after exhaustive exercise compared to CHO alone [30]. Although caffeine is highly effective in improving exercise tolerance, prescription for concurrent athletes remains circumstantial, and there is insufficient evidence of a positive effect on recovery within or between concurrent training sessions. Moreover, clearance of caffeine from the circulation occurs after 3–6 h, and thus some concerns have been raised regarding the timing of consumption and the potential effect on sleep [27]. This is of particular relevance for athletes undertaking multiple sessions in 1 day, with sleep between sessions likely to be negatively affected. Despite extensive evidence of the ergogenic effects of caffeine for different modes of exercise, evidence for improved concurrent training is lacking. However, given the concerns of increased load from concurrent training, caffeine may provide some benefits to aid preparation for ensuing sessions.

Antioxidants

An acute bout of exercise represents a disruption to redox balance via a short-term elevation of reactive oxygen species (ROS) in the cells [31], whilst frequent training elicits adaptive responses by which the resistance to oxidative stress becomes greater over time [31]. Thus, despite detrimental cellular effects of ROS, regular pro-oxidative states may strengthen antioxidant mechanisms and allow adaptation to subsequent oxidative challenges [31]. Conversely, excessive ROS production (e.g. through intense/prolonged exercise) can cause muscle damage [32], soreness [31] and a localised decrement in force production [33]. In concurrent training where resistance loading may result in damage and aerobic exercise in metabolic challenges, the role of antioxidants may have merit. Thus, strategies to attenuate these changes via modulation of redox mechanisms have attracted some interest in the recovery literature. For example, antioxidant supplementation (vitamin E) is suggested to defend sarcolemmal membranes against ROS-induced destabilisation and creatine kinase release [34]. Further, there is some evidence to suggest that antioxidant treatments can reduce fatigue-induced decrements in muscle contractile force [35]. However, current evidence for an effect of antioxidants to improve recovery, soreness or delayed onset muscle soreness (DOMS) is equivocal [31] and justification for use during concurrent training remains conceptual only [36]. Moreover, it has been observed that supplementary antioxidants (vitamin C) may blunt the exercise-induced spike in the expression of cytoprotective proteins, i.e. interfere with positive cellular adaptations [37]. In summary, although antioxidant supplementation may reduce the acute oxidative impact of strenuous exercise, it is

likely inappropriate as a long-term recovery strategy. Although a number of commercially available supplements have yet to be formally investigated, a lack of evidence specific to concurrent training renders their use unsubstantiated, and for concurrent athletes, a sufficient dietary intake of adequate fruits and vegetables remains the primary focus [31].

Anti-inflammatories

The period following a concurrent exercise bout is characterised by a localised elevation in leukocytes and inflammatory cytokines via an integrated tissue remodelling response [38, 39]. Specifically, it is suggested that macrophage-based prostaglandin synthesis, as well as localised oedema, stimulates pain receptors and contribute to feelings of DOMS [40]. Accordingly, for concurrent athletes seeking to undertake consecutive exercise bouts with abbreviated recovery periods, inflammatory processes have become a target for post-training interventions [41]. Purportedly, pharmacological and nutraceutical strategies may reduce training-induced performance decrements by manipulating inflammatory pathways and the associated neuromuscular mechanisms [42, 43]. However, some concerns are present with this strategy, as inhibition of inflammatory processes may interfere with important adaptation and repair mechanisms in skeletal muscle [44]. Thus, these interventions are dependent on contextual factors and should be interpreted with an understanding of the appropriateness of their application.

A number of foods and medicines are recognised to have anti-inflammatory properties, and thus there has been some investigation into supplementation as a means of blunting the inflammatory response to exercise. For example, curcumin (a flavonoid present in turmeric) may exert an anti-inflammatory effect via multiple mechanisms including reduced NF- κ B activity and cytokine release [45] and improved scavenging of free radicals [46]. Despite numerous data showing an anti-inflammatory effect with curcumin, there is a lack of research specific to concurrent training recovery. A study in mice showed that oral administration of curcumin prior to eccentric running attenuated the localised inflammation and muscle damage present with a placebo group, and also reversed training-induced performance decrements for 3 days [46]. Further, a pre- and post-exercise curcumin supplement caused a reduction in DOMS, inflammatory biomarkers, and improved jumping performance 24 h after eccentric resistance exercise in healthy men [47]. Indeed, there is preliminary support for the use of curcumin to augment recovery from exercise-induced muscle damage [48]. However, many supporting studies utilise eccentric-based exercise to induce DOMS and inflammatory responses, making findings difficult to apply to concurrent resistance and endurance training [48]. Alternatively, pharmacological anti-inflammatory treatments are commonly used amongst athletes due to their proposed anti-inflammatory and analgesic effects [49], which may aid the recovery process.

Non-steroidal anti-inflammatory drugs (NSAIDs), such as ibuprofen and aspirin, have been suggested as a potential recovery tool. These drugs work to reduce

inflammation via inhibition of cyclooxygenase-2 (COX-2) activity and subsequent prostaglandin synthesis [46]. If inflammation is considered a primary driver of DOMS and reduced muscle function, NSAIDs could conceivably play a role in minimising these effects. Previous investigations of this strategy have provided underwhelming results. In an acute setting, an ibuprofen tablet every 8 h post-eccentric resistance exercise caused reduced muscle damage markers, and a non-significant decrease in 48 h muscle soreness, but did not affect the acute recovery of strength/power performance in healthy participants [42]. Separately, untrained subjects showed no improvement in DOMS or strength performance after an eccentric exercise bout with NSAID treatment, and this outcome did not change for 3 days post-exercise [50]. There is some evidence to suggest that prophylactic NSAID administration (i.e. before exercise) may be more effective to improve recovery versus a post-exercise dose [51]; however, evidence in concurrent training models is lacking. Whilst promising evidence for the use of some anti-inflammatory compounds exists to reduce acute inflammation and possibly aid recovery, however, due to a lack of long-term studies relevant to concurrent training, it is difficult to make specific recommendations for such athletes.

Sleep and Timing of Recovery Periods

Sleep is an altered state of consciousness, without loss of neurophysiological control, and given the large periods in which sleep is undertaken, it represents a concentrated recovery period where metabolic and neural adaptations occur [52]. Given the large daily and microcycle volumes required for concurrent training, alongside potential molecular crosstalk adaptations and perceived fatigue, sleep and recovery periods are important to promote ensuing recovery [53, 54]. Of concern is when athletes do not gain sufficient sleep, either due to sleep deprivation or sleep restriction [55]. Sleep deprivation refers to severe sleep loss (i.e. entire nights) [56], and negatively impacts recovery—although its occurrence in athletes seems minimal [56, 57]. Alternatively, sleep restriction involves the disturbance of normal sleeping patterns, and smaller disturbances of sleep (i.e. 50% normal duration) [58]. Previous research has shown greater negative effects of sleep deprivation than restriction on recovery, though neither are ideal for concurrent athletes [56]. For example, sleep restriction (50% normal sleep) after high-intensity interval training increased feelings of sleepiness, and decreased peak power output and exercise motivation compared to normal sleep [58]. Given that high-intensity aerobic training is a common component of concurrent training [59], the decrease in mental and physical recovery may hinder an athlete's capacity to perform subsequent resistance training [60], in turn reducing training quality and potential adaptations. Additional research showed that total sleep deprivation decreased sprint times, perceived liveliness, muscle glycogen content, and peak voluntary force and voluntary activation of the quadriceps compared to normal sleep [57]. From the standpoint of a concurrent athlete, such alterations following training highlight the negative impact that sleep deprivation may have on ensuing training. Overall, maintaining normal sleep

patterns (7–9 h/night) appears vital to prevent the negative effect of sleep restriction or deprivation on strength and power outcomes, and psychological recovery [56] and as such, may support next-day training sessions, and concomitant physiological responses. As a recommendation, athletes should consider appropriate sleep hygiene practices to aid sleep, including avoiding electronic equipment and bright light prior to bed time, and avoiding use of caffeine and alcohol after 3:00 pm [61]. Shorter duration afternoon naps (10–30 min) between training sessions may also be utilised to attenuate the negative effects of sleep loss on recovery, if not ensure improved readiness to train regardless of prior sleep debt [61]. However, further research into the effect of chronic sleep loss on adaptations to concurrent training is required so that more specific recommendations for recovery can be made [56].

Given concurrent training increases the number and volume of training sessions, appropriate recovery time becomes an important consideration. Between-session recovery periods, as well as the timing of sessions on a given day, can influence the cumulative fatigue that is evident with concurrent training and, in turn, affect training performance [62]. As evidence, Robineau et al. [62] investigated the effects of different recovery periods on strength and endurance measures over a 7-week concurrent training programme in rugby players. The authors reported that strength and endurance measures increased to a greater extent when 24 h recovery was provided between endurance and strength training sessions, as compared to 0 h and 6 h between sessions [62]. Additionally, Sale et al. [63] studied the effect of 20 weeks' endurance and resistance training performed on the same day, compared to separate days, in untrained men. Separate-day training increased maximal leg press strength compared to same-day training. However, measures of muscle size and endurance capacity were similar between the respective methods [63]. Seemingly, sufficient recovery periods between endurance and strength sessions are important to facilitate adaptations from concurrent training; however, the abovementioned studies did not involve highly trained athletes, who may have greater training demands, and limitations on between-session recovery [64].

Many elite athletes undertake separate sessions of resistance and aerobic exercise in the same day (e.g. morning and evening). In this context, the recovery time between sessions, as well as the session order (i.e. endurance then resistance or vice versa) are worthy of consideration. Current evidence indicates that prior endurance exercise can negatively affect subsequent resistance training via inhibited force production [65] and a downregulated protein signalling response [4]. Further, it is suggested that this suppression of hypertrophic adaptations persists for at least 6 h [66], and possibly as long as 8 h [9], following an endurance-based training session. Thus it is suggested that endurance exercise be performed earlier in the day, with ≥ 6 h recovery before the later resistance training session [4]. As further evidence, the metabolic signalling response to aerobic exercise is reported to persist for 3 h post-session [67], and thus delaying resistance training until later in the day may permit sufficient recovery to allow the hypertrophic signalling cascade to proceed without interference [7]. An additional rationale for this strategy pertains to 'nutritional recovery', in that sufficient recovery time between same-day sessions also allows the athlete time to replenish protein and CHO [7]. The combined benefit of longer

durations between sessions and adequate nutritional intake ensures resistance training is performed in an energy-rich state, maximising protein signalling responses to the session and reducing the blunting effect of glycogen depletion on protein synthetic signalling [15]. Notwithstanding specific recovery methods, it is apparent that exercise-induced fatigue precipitates a split-session approach whereby training is organised such that recovery periods are sufficient to minimise fatigue-induced performance decrements [68].

Other Recovery Strategies

Whilst training load periodisation and nutritional intake represent the biggest drivers of recovery for concurrent athletes, implementation of specific recovery methods to improve between-session recovery and subsequent performance are likely to be of importance. Cold-water immersion (CWI) is one such method that may be used to improve symptoms of DOMS [69], and reduce performance decrements following exercise [70]. This strategy is suggested to ameliorate the central fatigue associated with prolonged or intense exercise and reduce post-session decrements in muscle function [71]. Specifically, CWI has been shown to improve subjective ratings of muscle soreness following intermittent running [72], team sport [73] and eccentric resistance training [74]; and may help to reverse decrements in muscle function after high-intensity exercise [72, 75]. When concurrent athletes are required to perform an increased number of sessions per day or within a microcycle, alleviation of DOMS may be important to ensure readiness to train and the quality of work performed.

Despite potential as an acute recovery strategy, the efficacy of CWI as a long-term recovery method has been questioned [76]. Recently, studies have suggested that the use of CWI after resistance training may inhibit strength and hypertrophic gains, with recent data showing blunted resistance training adaptations with post-exercise CWI [76, 77]. Acute studies support this notion, showing that post-resistance exercise CWI may attenuate muscle protein synthesis signalling (p70S6k), supposedly via reduced blood flow to skeletal muscle [77]. Further, the specific cooling protocol used seems to influence the response, with prolonged CWI inhibiting glycogen resynthesis, although current research is inconclusive on this matter [78]. In brief, despite showing promise as a tool to restore acute performance decrements, CWI may interfere with training adaptations (especially resistance training) when applied regularly. As such, care should be taken to consider the appropriateness of this intervention to the concurrent athlete.

The use of compression garments (CGs) may also facilitate recovery from concurrent training via reduced muscle damage and post-exercise ratings of soreness [79]. One suggested mechanism for this effect is compression of diluted blood vessels, leading to reduced levels of oedema and cell trauma [80]. Minimising residual fatigue from training is vital to maximise an individual's capacity to complete and adapt to concurrent training; hence, compression garments applied to fatigued muscle groups may be effective to restore the performance decrements associated with

successive sessions. As evidence, an accelerated recovery of strength performance compared to a non-CG control has been reported [81], though the explanatory mechanisms were unclear as muscle damage markers (myoglobin, creatine kinase) were similar across the two groups. Conversely, the effect on recovery from endurance exercise is less clear. A recent meta-analysis showed no benefit to post-running CGs, and the authors surmised that this was due to the multi-factorial nature of exercise-induced fatigue, whereby interventions to reduce muscle damage may have a minimal effect on overall recovery of performance [79]. During concurrent training, however, resistance training seems to have no detrimental effects on endurance training outcomes [5] and as such the necessity of CGs is questionable in this scenario. Moreover, a key concurrent recovery consideration is endurance exercise-induced fatigue, and presently, there are insufficient data to recommend CGs as a recovery strategy for the concurrent athlete. However, the use of CGs is regularly reported to improve perceived recovery, often via a placebo effect, and such perceptual assistance may still be of use for concurrent training to facilitate perceived readiness to train.

Summary

During concurrent training, appropriate recovery strategies are critical to maximise the individual's capacity to complete sessions, avoid injury and optimise adaptations. Generally, concurrent training presents a range of methodological obstacles relating to session schedules, rest periods and nutritional intake. Accordingly, an individualised approach targeting load distribution, nutrient timing and potential novel recovery interventions seems to be most appropriate for the concurrent athlete. Given the range and diversity of potential recovery interventions, Table 15.1 highlights a selection of these interventions and associated ranking of evidence to support their use. Future research should further examine the role of specific recovery methods within the context of concurrent training.

Practical Applications

- Fundamental recovery strategies (load scheduling, sleep and protein/carbohydrate consumption) should be the primary methods used by athletes undertaking concurrent training.
- Increased fatigue induced by large concurrent training volumes may be partially alleviated with the use of ergogenic aids such as creatine, nitrate and caffeine; though further mode-specific evidence is required.
- Nutraceutical and pharmacological recovery aids such as antioxidants and anti-inflammatories show promise to reduce acute fatigue. However, their long-term applicability and possible interference effects raise questions relating to potential applications for concurrent athletes.

- Compression garments and water immersion techniques are effective to enhance perceived recovery from muscle-damaging interventions (e.g. resistance training), although the effect on endurance performance recovery is less clear.
- Ensuring adequate rest must be a primary consideration for concurrent athletes—allowing at least 6 h recovery time between training sessions allows individuals to minimise fatigue-induced performance decrements, and provides time to implement nutrition-based recovery strategies. Athletes should aim for 7–9 h sleep per night, and follow appropriate sleep hygiene practices to minimise disturbances.

Table 15.1 Levels of evidence for a variety of potential recovery interventions for athletes undertaking concurrent training

Level of evidence	Intervention	Dosage	Proposed effect
A Body of evidence can be trusted to guide practice	Protein	1.7–2.2 g kg day ⁻¹	Aid muscle remodelling and promote strength adaptations
	Carbohydrate	3–12 g kg day (depending on activity)	Restore hepatic and intramuscular glycogen; support immune function
	Creatine	~5 g day in 3–5 day cycle	Improve PCr synthesis and buffer against acidosis
B Body of evidence can be trusted to guide practice in most situations	Caffeine	3–10 mg kg ⁻¹ (depending on individual response to dose)	Reduce perceptions of fatigue
C Body of evidence provides some support for recommendation(s) but care should be taken in its application	Cold-water immersion	10–15 min, 10–15 °C	Reduce soreness and restore muscle function via reduced central fatigue
	Curcumin	~25 g day ⁻¹	Reduce DOMS and muscle damage via anti-inflammatory effects
	Nitrate	400–600 mg day ⁻¹	Improve blood flow, mitochondrial respiration and muscle contractility
D Body of evidence is weak and recommendation must be applied with caution	Antioxidants		Protect against oxidative stress and reduce acute fatigue
	Compression garments		Reduce oedema and DOMS
	NSAIDs	~1200 mg day ⁻¹ (400 mg initial dose)	Reduce DOMS and muscle damage via anti-inflammatory effects

NHMRC levels of evidence and grades for recommendations for developers of guidelines [82]

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